

Ethical Phishing Simulation Platform

Abstract

This project focuses on developing an Ethical Phishing Simulation Platform to educate users about phishing attacks and improve cybersecurity awareness. The system simulates phishing campaigns in a controlled environment, tracks user interactions such as link clicks and credential submissions, and provides analytical insights. The platform is built using Flask, SQLite, and SMTP-based email services, ensuring a lightweight and scalable solution for training purposes.

Introduction

Phishing is one of the most common social engineering attacks used by cybercriminals to steal sensitive information. Organizations often conduct phishing simulations to train employees to recognize and avoid such attacks. This project aims to build a web-based phishing simulation tool that mimics real-world phishing scenarios in a safe and ethical manner. It enables administrators to send simulated phishing emails, track user responses, and generate insights to enhance security awareness.

Tools Used

- **Python:** Core programming language
- **Flask:** Web framework for backend development
- **SQLite:** Lightweight database for storing tracking data
- **HTML/CSS:** Frontend design for phishing pages and dashboard
- **SMTP (Gmail):** Email sending mechanism
- **Visual Studio Code:** Development environment

Steps Involved in Building the Project

1. Set up the development environment using Python and Flask.
2. Designed the database schema to store user interaction data.
3. Created phishing email templates and a login simulation page.
4. Implemented email sending functionality using SMTP.

5. Developed routes to track clicks and form submissions.
6. Built an analytics dashboard to display campaign results.
7. Added logging for monitoring user activities.
8. Tested the system in a controlled environment to ensure accuracy and safety.

Conclusion

The Ethical Phishing Simulation Platform successfully demonstrates how phishing attacks operate while providing a safe learning environment. It helps users identify potential threats and improves cybersecurity awareness. The project can be further enhanced by adding advanced analytics, user authentication, and real-time monitoring features, making it suitable for enterprise-level training programs.